



Martin Fournier

Computational Astrophysicist

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0009-0006-2593-1583

RESEARCH INTERESTS Intracuster medium · Computational astrophysics · Astrophysical fluid dynamics

ACADEMIC POSITIONS

2026 – present **Postdoctoral Researcher** Paris, France
Institut d'Astrophysique de Paris (IAP) · CNRS & Sorbonne Université · since October 2026

EDUCATION

2023 – 2026 **PhD in Astrophysics (Dr. rer. nat.), summa cum laude** Hamburg, Germany
Universität Hamburg
Thesis “Cold gas in a hot universe: Supermassive black hole feedback and multiphase condensation in the magnetized intracluster medium”
Advisors Prof. Marcus Brüggen & Dr. Philipp Grete

2022 – 2023 **MSc in Astronomy & Astrophysics** Paris, France
Observatoire de Paris & Université Paris Cité
Thesis “Cosmic-ray propagation in the Milky Way”, ENS Lyon
Advisors Dr. Jérémy Fensch & Dr. Benoît Commerçon

2019 – 2022 **MSc in Physics & Engineering** Grenoble, France
Phelma · Grenoble INP · Université Grenoble Alpes
Internship “Properties of Magellanic Cloud analogues in TNG50”, Max-Planck-Institut für Astronomie
Advisor Dr. Annalisa Pillepich

PUBLICATIONS

4 first-author and 7 co-authored papers · A&A, ApJ, MNRAS

FIRST AUTHOR

- 2026 XMAGNET – Stir before serving: a Lagrangian perspective on mixing-driven condensation in the intracluster medium. **M. Fournier** et al. · A&A, accepted · [1 citation](#)
- 2025 XMAGNET: Velocity structure functions of active galactic nucleus-driven turbulence in the multiphase intracluster medium. **M. Fournier** et al. · A&A · [10 citations](#)
- 2024 The properties of magnetised cold filaments in a cool-core galaxy cluster. **M. Fournier** et al. · A&A · [23 citations](#)
- 2023 Past activity of Sgr A^{*} is unlikely to affect the local cosmic-ray spectrum. **M. Fournier** et al. · A&A

CO-AUTHOR

- 2026 BlackHoleWeather – Spin-coupled chaotic cold accretion across the meso scale: morphology and thermodynamics. O. Piana, ..., **M. Fournier** et al. · submitted to A&A
- 2026 BlackHoleWeather – Jet-regulated chaotic cold accretion across the meso scale: morphology and thermodynamics. V. Cammelli, ..., **M. Fournier** et al. · submitted to A&A
- 2026 BlackHoleWeather – Chaotic cold accretion across the meso-scale: variability and kinematics. F. Barbani, ..., **M. Fournier** et al. · submitted to A&A
- 2026 BlackHoleWeather – Chaotic cold accretion across the meso-scale: morphology and thermodynamics. F. Barbani, ..., **M. Fournier** et al. · A&A, accepted

2025	XMAGNET: Kinetic, thermal and magnetic AGN feedback in massive galaxies at halo masses $\sim 10^{13.5} M_{\odot}$. D. Prasad, ..., M. Fournier et al. · MNRAS
2025	The XMAGNET exascale MHD simulations of SMBH feedback in galaxy groups and clusters: overview and preliminary cluster results. P. Grete, ..., M. Fournier et al. · ApJ · 8 citations
2024	Milky Way and Andromeda analogues from the TNG50 simulation. A. Pillepich, ..., M. Fournier et al. · MNRAS · 111 citations

COMPUTING TIME ALLOCATIONS

2024 – 2027	Co-PI <i>Magnetohydrodynamical simulations of jet–cloud interactions in the intracluster medium</i> 3 M core-h on JUPITER GPU Booster, Jülich Supercomputing Centre2026 – 2027 2 M core-h on JUWELS GPU Booster, Jülich Supercomputing Centre2024 – 2026
2023 – 2026	Contributor <i>Feedback and energetics from magnetized AGN jets in galaxy groups and clusters</i> 80 M core-h on Frontier, Oak Ridge Leadership Computing Facility2023 – 2026

TALKS

INVITED

2026	Parthenon User Meeting	Boston (online)
2025	BlackHoleWeather: Toward a Unified Theory of Feeding and Feedback	Sexten

CONTRIBUTED

2026	8th ICM Theory and Computation Workshop (poster talk)	Brno Observatory & Planetarium
2025	Modelling of Multiphase Astrophysical Media	Schloss Ringberg
2025	MIST2025 – Cosmic Turbulence and Magnetic Fields	Cargèse
2025	Journées de la Société Française d’Astronomie & d’Astrophysique (SF2A)	Toulouse
2024	Galaxy Clusters & Radio Relics II	Cambridge, MA
2024	Journées de la Société Française d’Astronomie & d’Astrophysique (SF2A)	Marseille

SEMINARS & COLLOQUIA

2024 – 2026	Harvard–Smithsonian CfA (2024, 2025) · Observatoire de Paris (2025, 2026) · Princeton University (2025) Max-Planck-Institut für Meteorologie (2025) · Newcastle University (2024) Institut für Theoretische Astrophysik (2024) · Max-Planck-Institut für Astronomie (2024) · ENS Lyon (2024)
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TEACHING & SERVICE

Teaching	Lecturer <i>Introductory Python for Physicists</i> Universität Hamburg · 23 h Teaching assistant <i>Active Galactic Nuclei and Supermassive Black Holes</i> Universität Hamburg · 12 h
Refereeing	Astronomy & Astrophysics (A&A)

OUTREACH

Organiser	Main organiser of a sold-out Science Slam show at the Hamburg Planetarium (250+ attendees).
Speaker	Popular-science talks and Science Slams at various public outreach events.
Visuals	Production of videos based on the IllustrisTNG simulations, featured on ARTE and in <i>When Fish Began to Crawl</i> (2023), a film on climate change by BAFTA-winning director Morgan McKinnon, premiered at the Edinburgh Science Festival.

TECHNICAL SKILLS

Languages	Python · C++ · Fortran · $\LaTeX$
Codes	ATHENAPK · KOKKOS · AREPO · RAMSES